

By Mahar Afaq Safdar Muhammadi

MTH644 Current Papers Questions (Fall 2021 Final Term)

① If f is measurable and $k > 0$ then show that kf is measurable. ~~$\forall k \in \mathbb{R}$~~

Sol: " f " is measurable.

$$\Rightarrow \{x: f(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}$$

$$\text{If } k > 0, \therefore f(x) > a, \forall a \in \mathbb{R}$$

$$\Rightarrow f(x) > \frac{a}{k} \Rightarrow kf(x) > a, \forall a \in \mathbb{R}$$

$$\therefore \{x: kf(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}$$

So " kf " is measurable.

② If f, g and $f+g$ is measurable then $f-g$ is measurable.

Sol: Given that f, g and $f+g$ is measurable.

We have to prove " $f-g$ " is measurable.

$$\because g \text{ is measurable} \Rightarrow (-1)g = -g \text{ is also } \mathcal{M}$$

$$\Rightarrow f + (-1)g = f - g$$

$$\because f+g \text{ is measurable} \Rightarrow f-g \text{ is also measurable.}$$

③ If $m(A)$ and $m(B)$ are ^{measurable} infinity then prove $m(A) + m(B) = m(A \cup B) + m(A \cap B)$.

Sol: Given that $m(A) = m(B) = \infty$, then result is obvious.

$$\Rightarrow m(A \cup B) + m(A \cap B) = \infty,$$

$$\text{So } m(A) + m(B) = m(A \cup B) + m(A \cap B)$$

(proved)

(4) Find dropped interval in cantor set, for $n=1$

Sol: We know that

$$m = 2^{n-1}$$

$$\text{for } n=1, \quad m = 2^{1-1} = 2^0 = 1$$

Hence dropped interval G_1 is

$$G_1 = I_{mm} = I_{11}$$

(5) Define Borel Measure Space.

Sol: Restriction of Lebesgue measure "m" to the class of Borel set "B" is said to be Borel measure space $(\mathbb{R}, B, m)$.

(6) $f(x) = |x+2|$, find f^+ for $f(x) \geq 0$

Sol: We know that

$$f^+ = \frac{f + |f|}{2}$$

$$f = |x+2| \quad \text{and} \Rightarrow f = x+2 \quad \because f(x) \geq 0$$

$$|f| = ||x+2|| \Rightarrow |f| = x+2 \quad \because f(x) \geq 0$$

So

$$f^+ = \frac{x+2 + x+2}{2} = \frac{2x+4}{2}$$

$$f^+ = \frac{2(x+2)}{2}$$

$$f^+ = x+2 \quad \underline{\text{Ans.}}$$

$\exists$ there exists

⑦ Lebesgue measure m is translation invariant.

sol: Given that say $\phi \neq E \in \mathcal{M}$ (measurable)

$\therefore E$ is measurable $\Leftrightarrow$ for given $\epsilon > 0$, $\exists$ an open set $U \supseteq E$ such that $m^*(U-E) < \epsilon$

" U " is open set in $\mathbb{R} \Rightarrow "U+y"$ is also open in $\mathbb{R}$.
 $\forall y \in \mathbb{R}$

If $E \subseteq U \Rightarrow E+y \subseteq U+y$

$$\Rightarrow (U+y) - (E+y) = (U-E) + y$$

$$m^*[(U+y) - (E+y)] = m^*[(U-E) + y] = m^*(U-E) < \epsilon$$

$$\text{But } m^* = m \Rightarrow m[(U+y) - (E+y)] = m(U-E) < \epsilon$$

$\Rightarrow "E+y"$ is also measurable.

$\therefore "m"$ is translation invariant. (proved)

⑧ Let $E \subseteq A = [0, 1]$ be a measurable set, then for each $y \in A$. The set $E \hat{+} y$ is measurable and $m(E) = m(E \hat{+} y)$.

Sol: Suppose $E_1 = E \cap [0, 1-y]$ and $E_2 = E \cap [1-y, 1]$

Then set $E_1, E \cap [0, 1-y]$ and $E \cap [1-y, 1]$ are measurable

So E_1 and E_2 are also measurable. Further

$$E_1 \cup E_2 = E \text{ and } E_1 \cap E_2 = \phi$$

$$\Rightarrow m(E) = m(E_1) + m(E_2) \rightarrow \textcircled{1} "m" \text{ is finitely additive.}$$

Let $y_1 \in E_1$ then $y_1 < 1-y \Rightarrow y_1 + y < 1$

$$\Rightarrow y_1 \hat{+} y = y_1 + y \Rightarrow E_1 \hat{+} y = E_1 + y \rightarrow \textcircled{2}$$

Now if $y_2 \in E_2 \Rightarrow y_2 \geq 1-y \Rightarrow y_1 + y_2 \geq 1$

$$\Rightarrow y_2 \hat{+} y = y_2 + y - 1 \Rightarrow E_2 \hat{+} y = E_2 + (y-1) \rightarrow (b)$$

" m is translation invariant.

$\Rightarrow E_1 + y$ and $E_2 + (y-1)$ are measurable.

$\Rightarrow E_1 \hat{+} y$ and $E_2 \hat{+} (y-1)$ are measurable

$$\Rightarrow \left. \begin{aligned} m(E_1 \hat{+} y) &= m(E_1 + y) = m(E_1) \\ \text{and } m(E_2 \hat{+} y) &= m(E_2 + (y-1)) = m(E_2) \end{aligned} \right\} \rightarrow (2)$$

" $E_1 \cup E_2 = E$ and $E_1 \cap E_2 = \phi$

$$\Rightarrow E \hat{+} y = (E_1 \hat{+} y) \cup (E_2 \hat{+} y) \text{ and } (E_1 \hat{+} y) \cap (E_2 \hat{+} y) = \phi$$

$$\Rightarrow m(E \hat{+} y) = m(E_1 \hat{+} y) + m(E_2 \hat{+} y) \quad (\text{union of two disjoint measurable sets})$$

$$= m(E_1) + m(E_2) \quad \text{by (2)}$$

$$m(E \hat{+} y) = m(E) \quad \text{by (1) (proved)}$$

(9) A subset $\phi \neq E \subseteq \mathbb{R}$ is measurable $\Leftrightarrow$ for given $\epsilon > 0 \exists$ an open set $U \supseteq E$ with $m^*(U-E) < \epsilon$ (finite)

Sol: Suppose E is measurable.

If $m^*(E) < \infty$, $\forall \phi \neq E \subseteq \mathbb{R}$, for $\epsilon > 0$, $\exists$ an open set $U \supseteq E$, $\Rightarrow m^*(U) < m^*(E) + \epsilon \rightarrow (a)$

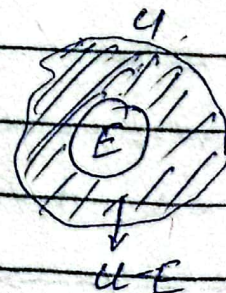
$$E \subseteq U \Rightarrow U = E \cup (U-E)$$

$$\Rightarrow m^*(U) = m^*(E) + m^*(U-E)$$

$$\Rightarrow m^*(U-E) = m^*(U) - m^*(E) < \epsilon$$

$$\Rightarrow m^*(U-E) < \epsilon \quad (\text{from eq (a)})$$

(proved)



(10) Prove that every Borel set is measurable.

Sol: Every open and closed set is measurable.

$\Rightarrow$ countable union or intersection of closed or open set is also measurable.

Therefore Borel set is also measurable being union or intersection of closed or open sets.

(11) Give examples of Borel set.

Sol: Examples of Borel set:

(i) $G_8 = \bigcap_{n=1}^{\infty} U_n$

(ii) $F_8 = \bigcup_{n=1}^{\infty} F_n$

(12) Let $\{f_n\}$ be a sequence of extended real valued measurable function on "D" then prove that $\sup_{n \in \mathbb{N}} (\inf_{k \geq n} f_k)$ is also measurable.

Sol: let $p = \sup_{n \in \mathbb{N}} (\inf_{k \geq n} f_k)$ and

$$G_n = \inf_{k \geq n} \{f_n, f_{n+1}, f_{n+2}, \dots, f_{n+k}, \dots\}, \forall n \in \mathbb{N}$$

Each G_n is measurable by the measurability of f_n for each n and $\inf_{n \in \mathbb{N}} f_n$ is measurable.

Consequently $p = \sup_{n \in \mathbb{N}} G_n$ is a measurable function because $\sup_{n \in \mathbb{N}} f_n$ is measurable.

(13) Let $f: E \rightarrow \mathbb{R} \cup \{\pm\infty\}$, $\phi \neq E \in \mathcal{M}$, where $\mathcal{M}$ be measurable space. Then show that:

$$\{x \in E: f(x) \leq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E: f(x) < a + \frac{1}{n}\}$$

Sol:

Let $y \in \{x \in E: f(x) \leq a\} \rightarrow (a)$

$$\Rightarrow f(y) \leq a, \forall a \in \mathbb{R}.$$

$$\Rightarrow f(y) < a + \frac{1}{n}, \forall n \in \mathbb{N}$$

$$\Rightarrow y \in \{x \in E: f(x) < a + \frac{1}{n}\}, \forall n \in \mathbb{N}$$

$$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E: f(x) < a + \frac{1}{n}\} \rightarrow (b)$$

From eq. (a) and (b)

$$\{x \in E: f(x) \leq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E: f(x) < a + \frac{1}{n}\} \quad (\text{proved})$$

(14) Evaluate $m(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm})$ where interval I_{nm} are pairwise disjoint and $m(I_{nm}) = \mu(I_{nm}) = \frac{1}{3^n}$

Sol:

$$\text{Let } m(G) = m(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm})$$

$$= \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} m(I_{nm}) = \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} \mu(I_{nm})$$

$$= \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} \left(\frac{1}{3^n}\right)$$

$$= \sum_{m=1}^{2^1-1} \left(\frac{1}{3^1}\right) + \sum_{m=1}^{2^2-1} \left(\frac{1}{3^2}\right) + \sum_{m=1}^{2^3-1} \left(\frac{1}{3^3}\right) + \dots$$

$$m(G) = \frac{1}{3} + \left(\frac{1}{3^2} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3}\right) + \dots$$

$$m(G) = \frac{1}{3} + \frac{2}{3^2} + \frac{4}{3^3} + \dots$$

$$m(G) = 1$$

$$a = \frac{1}{3}, r = \frac{2}{3}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{\frac{1}{3}}{1-\frac{2}{3}}$$

$$S_{\infty} = \frac{\frac{1}{3}}{\frac{1}{3}} = 1$$

$$m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right) = 1 \quad \underline{\underline{\text{Ans}}}$$

(15) Check the measurability of the cantor set in n th steps.

Sol: In general, at n th step, 2^{n-1} intervals I_{nm} for $m=1, 2, 3, \dots, 2^{n-1}$ are removed. Each I_{nm} has obviously length equal $\frac{1}{3^n} = 3^{-n}$. We have a

Sequence $\{I_{nm}\}$ of pairwise disjoint open sets.

Thus the set $G = \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}$ being the union of a sequence of open sets is open and measurable. The complement of G with respect to $[0, 1]$ is called cantor set and is denoted by C . The set C is closed and so it must be a measurable.

(16) If "f" and "g" are two ^{extended} real valued and measurable functions defined on same domain $\phi \neq \emptyset$ then prove that $f \wedge g$ is measurable.

Sol: $\because f \wedge g = \min(f, g) = \frac{1}{2}[(f+g) - |f-g|]$

So $f \wedge g$ is measurable because $(f+g)$ and $|f-g|$ are measurable and 'kf' is measurable $\forall k \in \mathbb{R}$.

OR

$$(f \wedge g)(x) = \min(f(x), g(x))$$

observe that any $a \in \mathbb{R}$, we have

$$\{x: (f \wedge g)(x) > a\} = \{x: f(x) > a\} \cap \{x: g(x) > a\}$$

$f(x)$ and $g(x)$ are measurable functions and the intersection of two measurable function is also measurable $\Rightarrow (f \wedge g)(x)$ is measurable.

(17) Determine the measure 1st step construction of cantor set from $[0, 1]$

Sol: We know that

$$m(G) = m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right)$$

For 1st step

$$\text{put } n=1 \Rightarrow m = 2^{n-1} = 2^{1-1} = 2^0 = 1$$

$$m(G) = m(I_1) = l(I_1)$$

$$m(G) = m(I_1) = l(I_1) = \frac{1}{3} = \frac{1}{3} \quad \because l(I_{nm}) = \frac{1}{3^n}$$

Ans

(18) Let $f: E \rightarrow \mathbb{R} \cup \{\pm\infty\}$ and $\phi \neq E \in \mathcal{M}$, be the measurable space then prove that $\{x \in E: f(x) > 0\}$ is also measurable

Sol: By the definition of measurable function $f: E \rightarrow \mathbb{R}$ is Lebesgue measurable function on $\phi \neq E \in \mathcal{M}$ if $\{x \in E: f(x) > a\} \in \mathcal{M}$ (measurable) for every $a \in \mathbb{R}$.

So $\{x \in E: f(x) > 0\}$ is also measurable because $0 \in \mathbb{R}$.

(19) Let $\phi \neq E \subset \mathbb{R}$, show that there exist F_0 set $F \subset E$.

Sol: For each n , let F_n be a closed set with $F_n \subset E$ and $m^*(E - F_n) < \frac{1}{n}$. Then $F = \bigcup_{n=1}^{\infty} F_n$ is an F_σ set. $F_n \subset \bigcup_{n=1}^{\infty} F_n = F \subset E$ proved

(20) If $\phi = A, B \subset X$ and same member x are in $A \cup B$ Then verify that $\chi_{A \cup B} = \chi_A + \chi_B - \chi_{A \cap B}$

Sol: Given that $A, B = \phi \Rightarrow \chi_A = \chi_B = 0$

$\Rightarrow A \cup B = \phi$ and $A \cap B = \phi \Rightarrow \chi_{A \cup B} = 0 \neq \chi_{A \cap B} = 0$

Hence prove that

$$\chi_{A \cup B} = 0 = \chi_A + \chi_B - \chi_{A \cap B}$$

(20) Let $\phi \neq E \subseteq \mathbb{R}$, if given that ① G_1 set " G " with $E \subseteq G$ and $m^*(G-E) = 0$ then prove that

② F_1 set " F " with $F \subseteq E$ and $m^*(E-F) = 0$

Sol: ① $\Rightarrow$ ② $\because \phi \neq E \subseteq \mathbb{R} \Rightarrow \phi = E' \subseteq \mathbb{R}$

$\therefore$ For $\phi \neq E'$ given $E' \subseteq G$, $m^*(G-E') = 0$

$$\Rightarrow m^*(G \cap (E')') = 0$$

$$\Rightarrow m^*(G \cap E) = 0 \Rightarrow m^*(E \cap G) = 0$$

$$\Rightarrow m^*(E \cap (G')') = 0 \Rightarrow m^*(E - G') = 0$$

$$\Rightarrow m^*(E - F) \text{ where } F \text{ is an } F_1 \text{ set}$$

being the complement of G_1 set G . i.e. $G' = F_1$

(21) Let $\phi \neq E$ be a measurable set, $f: E \rightarrow \mathbb{R} \cup \{\pm\infty\}$ then prove that $\{x \in E: f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\}$, $\forall n \in \mathbb{N}$

Sol: Let $Z \in \{x \in E: f(x) = -\infty\}$

$$\Rightarrow Z \in \{x \in E: f(x) < -n\}, \forall n \in \mathbb{N}$$

$$\Rightarrow Z \in \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\}, "$$

$$\Rightarrow \{x \in E: f(x) = -\infty\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\} \rightarrow \text{①}$$

Now let $S \in \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\}$, $\forall n \in \mathbb{N}$

$$\Rightarrow S \in \{x \in E: f(x) < -n\}, "$$

$$\Rightarrow f(S) < -n \Rightarrow S \in \{x \in E: f(x) = -\infty\}$$

$$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\} \subseteq \{x \in E: f(x) = -\infty\} \rightarrow \text{②}$$

from ① and ②

$$\text{So, } \{x \in E: f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in E: f(x) < -n\} \text{ (proved)}$$

② If f and g are extended real valued ^{and measurable} function.
Then prove $\bigcup_{p \in \mathbb{Q}} [\{x: f(x) > p\} \cap \{x: a - g(x) < p\}] \subseteq \{x: (f+g)(x) > a\}$

Sol: Let $t \in \bigcup_{p \in \mathbb{Q}} [\{x: f(x) > p\} \cap \{x: a - g(x) < p\}]$

$\Rightarrow t \in \{x: f(x) > p\} \cap \{x: a - g(x) < p\}$ for some $p \in \mathbb{Q}$

$\Rightarrow t \in \{x: f(x) > p\}$ and $t \in \{x: a - g(x) < p\}$

$\Rightarrow f(t) > p$ and $a - g(t) < p \Rightarrow p > a - g(t)$

$\Rightarrow f(t) > p > a - g(t) \Rightarrow f(t) > a - g(t)$

$\Rightarrow f(t) + g(t) > a \Rightarrow (f+g)(t) > a$

$\Rightarrow t \in \{x: (f+g)(x) > a\}$

So $\bigcup_{p \in \mathbb{Q}} [\{x: f(x) > p\} \cap \{x: a - g(x) < p\}] \subseteq \{x: (f+g)(x) > a\}$
(proved)

③ If " f " is real valued and measurable function.

Then f^2 is also measurable.

Sol: We need to prove $\{x: f^2(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}$

$$\{x: f^2(x) > a\} = \begin{cases} \{x: f(x) > \sqrt{a}\} \cup \{x: f(x) < -\sqrt{a}\} & ; a > 0 \\ \text{Domain} & ; a < 0 \end{cases}$$

$\because f$ is measurable $\Rightarrow \{x: f(x) > \sqrt{a}\} \in \mathcal{M}$ and $\{x: f(x) < -\sqrt{a}\} \in \mathcal{M}$

$\Rightarrow \{x: f(x) > \sqrt{a}\} \cup \{x: f(x) < -\sqrt{a}\} \in \mathcal{M}$ and

Domain is given to be measurable. i.e. Domain $\in \mathcal{M}$

$\therefore \{x: f^2(x) > a\} \Rightarrow f^2$ is also measurable.

(24) If $\phi \neq A, B \subseteq X$ Then prove that $\chi_{A-B} = \chi_A - \chi_{A \cap B}$

Sol: Case I: If $x \in A-B$

$$\Rightarrow x \in A \text{ but } x \notin B, x \notin A \cap B \Rightarrow \chi_{A-B} = 1$$

$$\chi_A - \chi_{A \cap B} = 1 - 0 = 1 = \chi_{A-B}$$

Case II: If $x \notin A-B$

$$x \notin A-B = \begin{cases} x \notin A, x \notin B, x \notin A \cap B \rightarrow \textcircled{a} \\ x \notin A, x \in B, x \notin A \cap B \rightarrow \textcircled{b} \\ x \in A, x \in B, x \in A \cap B \rightarrow \textcircled{c} \end{cases}$$

$$\Rightarrow \chi_{A-B} = 0$$

$$\chi_A - \chi_{A \cap B} = \begin{cases} \textcircled{a} \Rightarrow 0 - 0 = 0 \\ \textcircled{b} \Rightarrow 0 - 0 = 0 \\ \textcircled{c} \Rightarrow 1 - 1 = 0 \end{cases} = \chi_{A-B}$$

$\therefore$ For all cases $\chi_{A-B} = \chi_A - \chi_{A \cap B}$ (proved)

(25) Let $f: E \rightarrow \mathbb{R} \cup \{\pm\infty\}$ and $\phi \neq E \in \mathcal{M}$, where $\mathcal{M}$ be the measurable space. Then show that

$$\{x \in E: f(x) \geq a\} \in \bigcap_{n=1}^{\infty} \{x \in E: f(x) \geq a - \frac{1}{n}\}, \forall n \in \mathbb{N}$$

Sol: let $y \in \{x \in E: f(x) \geq a\} \Rightarrow f(y) \geq a, \forall a \in \mathbb{R}$

$$\Rightarrow f(y) \geq a - \frac{1}{n}, \forall n \in \mathbb{N}$$

$$\Rightarrow y \in \{x \in E: f(x) \geq a - \frac{1}{n}\}, \forall n \in \mathbb{N}$$

$$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E: f(x) \geq a - \frac{1}{n}\}$$

$$\Rightarrow \{x \in E : f(x) \geq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) \geq a - \frac{1}{n}\} \quad (\text{proved})$$

(26) For a given $\phi \neq E \subseteq \mathbb{R}$ and $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $m^*(E - F) < \varepsilon$, then show that $\exists$ an open set satisfying $m^*(O - E) < \varepsilon$.

Sol: For $\phi \neq E \subseteq \mathbb{R} \Rightarrow E' \subseteq \mathbb{R}$, $\exists$ a closed set " F " and $\varepsilon > 0$ such that $F \subseteq E'$ and $m^*(E' - F) < \varepsilon \rightarrow (1)$

$$F \subseteq E' \Rightarrow E = E' \subseteq F' = O \quad (\text{open set}) \quad \text{say}$$

$$\Rightarrow m^*(O - E) = m^*(O \cap E') = m^*(E' \cap O)$$

$$\Rightarrow m^*(O - E) = m^*(F' - O') = m^*(E' - F) < \varepsilon \quad \text{from (1)}$$

$$\Rightarrow m^*(O - E) < \varepsilon \quad (\text{proved})$$

(27) If $m^*(G - E) = 0$, then show that $m^*(G - E^c) = 0$ (2)

Sol: $\because m^*(G - E) = 0$, $\phi \neq E \subseteq \mathbb{R} \Rightarrow \phi \neq E' \subseteq \mathbb{R}$

$\therefore$ for $\phi \neq E'$ gives $E' \subseteq G$

$\because E \subseteq G$

$$\text{Hence } m^*(G - E^c) = 0 \quad (\text{proved})$$

(28) Let f be an extended real valued measurable function on D and $\phi \neq A \subseteq D$. Then show that

$\{x \in A : f(x) > a\}$ is measurable, $\forall a \in \mathbb{R}$. (3)

~~Sol: For f is a measurable function. $f: D \rightarrow \mathbb{R} \cup \{\pm\infty\}$~~

For Solution See Question No. 36

Q(29) Let "f" be an extended real valued and measurable functions defined on some domain $\phi \neq \emptyset$ the prove that $|f|$ is measurable function.

Sol: $\because$ "f" is measurable $\Rightarrow \{x: f(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}$

To prove $|f|$ is measurable, we need to show that

$$\{x: |f|(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}$$

$$\{x: |f|(x) > a\} = \{x: |f(x)| > a\}$$

$$\{x: |f|(x) > a\} = \begin{cases} \{x: f(x) > a\} \cup \{x: f(x) < -a\}; & a > 0 \\ \emptyset & ; a \leq 0 \end{cases}$$

In all cases on R.H.S, we get a measurable sets.

$\Rightarrow \{x: |f|(x) > a\}$ is measurable $\forall a \in \mathbb{R}$

Therefore $|f|$ is also measurable.

Q(30) Countable set are Borel sets with zero measure.

Sol: Case-I Singleton

$$\forall a \in \mathbb{R} : \{a\} = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, a + \frac{1}{n}) ; n \in \mathbb{N}$$

$\therefore \{(a - \frac{1}{n}, a + \frac{1}{n})\}_{n=1}^{\infty}$ is decreasing sequence.

$$\text{with } m(E_n) = \frac{2}{n} < \infty \text{ where } E_n = (a - \frac{1}{n}, a + \frac{1}{n})$$

$$\Rightarrow m\{a\} = m\left\{\bigcap_{n=1}^{\infty} E_n\right\} = \lim_{n \rightarrow \infty} m(E_n) = \lim_{n \rightarrow \infty} \frac{2}{n}$$

$$\Rightarrow m\{a\} = \lim_{n \rightarrow \infty} \left(\frac{2}{n}\right) = 0$$

Case II: Finite set

$$\text{let } \{x_1, x_2, x_3, \dots, x_n\} = \bigcup_{i=1}^n \{x_i\}$$

$$\Rightarrow m\left(\bigcup_{i=1}^n \{x_i\}\right) = \sum_{i=1}^n m\{x_i\} = 0$$

Case III: Infinite set

$$\text{let } \{x_1, x_2, x_3, \dots\} = \bigcup_{i=1}^{\infty} \{x_i\}$$

$$\Rightarrow m\left(\bigcup_{i=1}^{\infty} \{x_i\}\right) = \sum_{i=1}^{\infty} m\{x_i\} = 0$$

(proved)

(31) Open interval $(a, b) \in \mathbb{R}$ is measurable.

Sol: $(-\infty, a)$ is measurable.

$\Rightarrow (a, \infty)^c = (-\infty, a]$ is also measurable.

and $(-\infty, b - \frac{1}{n}]$ is also measurable $\forall n \in \mathbb{N}, n \rightarrow \infty$

and $(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - \frac{1}{n}]$ is also measurable.

because being countable union of measurable set

$\Rightarrow (a, \infty) \cap (-\infty, b) = (a, b)$ is also measurable.

because being intersection of two measurable set.

(32) If $A = \bigcup_{n=1}^{\infty} A_n$ and $\{A_n\}_{n=1}^{\infty}$ is a sequence of pairwise disjoint subsets of X . Then show

$$\text{that } \mu_A = \sum_{n=1}^{\infty} \mu_{A_n}$$

Sol: Case I: If $x \in A \Rightarrow x \in \bigcup_{n=1}^{\infty} A_n$

$\Rightarrow x \in A_k$ exactly for $n=k$

$$x \notin \bigcup_{n=1}^{\infty} A_n = \{A_k\}$$

$$\Rightarrow \chi_{A_k}(x) = 1, \chi_{A_n}(x) = 0, \forall n \neq k$$

$$x \in A \Rightarrow \chi_A(x) = 1$$

$$\therefore \chi_A(x) = 1 = 0 + 0 + \dots + \underbrace{1}_{n=k} + 0 + 0 + \dots$$

$$\chi_A(x) = 1 = \sum_{n=1}^{\infty} \chi_{A_n}(x) \Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$$

Case II: $x \notin A = \bigcup_{n=1}^{\infty} A_n \Rightarrow x \notin A_n, \forall n \in \mathbb{N}$

$$\Rightarrow \chi_A(x) = 0 \quad | \quad \Rightarrow \chi_{A_n}(x) = 0, \forall n \in \mathbb{N}$$

$$\chi_A(x) = 0 + 0 + 0 + \dots + 0 + \dots = \sum_{n=1}^{\infty} \chi_{A_n}(x)$$

$$\Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n} \quad (\text{proved})$$

~~Example (33) let $\{E_n, \uparrow\}_{n=1}^{\infty}$ be an increasing sequence of measurable sets, then show that~~

~~(33) let $\{E_n, \uparrow\}_{n=1}^{\infty}$ be an increasing sequence of measurable sets, then show that~~

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) = m\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} m^*(E_n)$$

Sol: Given that $(E_n, \uparrow)$, such that

$\Rightarrow E_1 \subset E_2 \subset E_3 \subset \dots$ of measurable sets.

Defining: $F_1 = E_1, F_2 = E_2 - E_1, F_3 = E_3 - E_2, \dots$

$E_n = E_n - E_{n-1}, \dots$ where $\{E_n\}_{n=1}^{\infty}$ is a sequence of disjoint measurable sets

$$\Rightarrow \bigcup_{k=1}^n F_k = \bigcup_{k=1}^n E_k \Rightarrow \bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k$$

$$\therefore m^*\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} F_k\right)$$

$$m^*(\bigcup_{k=1}^{\infty} E_k) = \sum_{k=1}^{\infty} m(E_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n m(E_k)$$

$$= \lim_{n \rightarrow \infty} m(\bigcup_{k=1}^n E_k) = \lim_{n \rightarrow \infty} m(\bigcup_{k=1}^{\infty} E_k)$$

$$= \lim_{n \rightarrow \infty} m^*(E_n) \quad (\text{proved})$$

(34)

Let $\{E_n\}_{n=1}^{\infty}$ be a decreasing sequence of measurable sets such that $m(E_k) < \infty$ for some $k \in \mathbb{N}$.

Then shows that $m^*(\bigcap_{n=1}^{\infty} E_n) = m(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m^*(E_n)$

Sol: Let $m(E_1) < \infty$, say

$$\therefore \langle E_n \rangle \rightarrow E_1 \supseteq E_2 \supseteq E_3 \supseteq \dots$$

$$\Rightarrow \phi = E_1 - E_1 \subseteq E_1 - E_2 \subseteq E_1 - E_3 \subseteq \dots \subseteq E_1 - E_n \subseteq \dots$$

$$\Rightarrow m^*(\bigcup_{n=1}^{\infty} (E_1 - E_n)) = \lim_{n \rightarrow \infty} m^*(E_1 - E_n)$$

$$\bigcup_{n=1}^{\infty} (E_1 - E_n) = \bigcup_{n=1}^{\infty} (E_1 \cap E_n')$$

$$= (E_1 \cap E_1') \cup (E_1 \cap E_2') \cup \dots$$

$$= E_1 \cap [E_1' \cup E_2' \cup E_3' \cup \dots]$$

$$= E_1 \cap [E_1 \cap E_2' \cap E_3' \cap \dots]'$$

$$= E_1 \cap (\bigcap_{n=1}^{\infty} E_n)'$$

$$\bigcup_{n=1}^{\infty} (E_1 - E_n) = E_1 - (\bigcap_{n=1}^{\infty} E_n)$$

$$\lim_{n \rightarrow \infty} m^*(E_1 - E_n) = m^*(E_1 - \bigcap_{n=1}^{\infty} E_n)$$

$$\lim_{n \rightarrow \infty} m^*(E_1) - \lim_{n \rightarrow \infty} m^*(E_n) = m^*(E_1) - m^*(\bigcap_{n=1}^{\infty} E_n)$$

$$m^*(E_1) - \lim_{n \rightarrow \infty} m^*(E_n) = m^*(E_1) - m^*(\bigcap_{n=1}^{\infty} E_n)$$

$$\Rightarrow m^*(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m^*(E_n) \quad (\text{proved})$$

35) If "f" and "g" are two extended real valued and measurable functions defined on same domain $\phi \neq \emptyset$ then prove that $f \vee g = \max(f, g)$ is measurable.

Sol: $\because f \vee g = \max(f, g) = \frac{1}{2} [(f+g) + |f-g|]$

So $f \vee g$ is measurable because $(f+g)$ and $|f-g|$ are measurable and kf is measurable for all $k \in \mathbb{R}$.

OR

$$(f \vee g)(x) = \max(f(x), g(x))$$

For any $a \in \mathbb{R}$, we have

$$\{x: (f \vee g)(x) > a\} = \{x: f(x) > a\} \cup \{x: g(x) > a\}$$

$f(x)$ and $g(x)$ are measurable functions and the Union of two measurable functions is also measurable.

$\Rightarrow (f \vee g)(x)$ is measurable.

36) Let f be an extended real valued measurable function on D and $\phi \neq A \subseteq D$. Then show that restriction of "f" to A is also measurable and $A \in \mathcal{M}$.
 OR $\{x \in A: f_A(x) > a\}$ is measurable, $\forall a \in \mathbb{R}$.

Sol: $\because f$ is measurable

$$\Rightarrow \{x \in D: f(x) > a\} \in \mathcal{M}, \forall a \in \mathbb{R}.$$

To prove " f_A " is measurable we have

$$\{x \in A : f_A(x) > a\} \in \mathcal{M}$$

$$\text{Note: } \{x \in A : f_A(x) > a\} = \{x \in D : f(x) > a\} \cap A$$

On R.H.S both sets are measurable.

$\therefore$ Intersection of two measurable sets is measurable

$$\Rightarrow f_A \text{ is measurable; } \{x \in A : f_A(x) > a\} \in \mathcal{M} \quad (\text{proved})$$

$$(37) \text{ Show that } f = f^+ - f^-$$

$$\text{Sol: If } f(x) \geq 0 \Rightarrow f^+(x) = f(x), f^-(x) = 0$$

$$\therefore (f^+ - f^-)(x) = f^+(x) - f^-(x) = f(x) - 0 = f$$

$$\text{If } f(x) < 0 \Rightarrow f^+(x) = 0, f^-(x) = -f(x)$$

$$\therefore (f^+ - f^-)(x) = f^+(x) - f^-(x) = 0 - (-f(x)) = f$$

$$\Rightarrow f^+ - f^- = f \quad (\text{proved})$$

$$(38) \text{ Show that } f^+ = \frac{f + |f|}{2}$$

$$\text{Sol: If } f(x) \geq 0 \Rightarrow f^+(x) = f(x) \Rightarrow f^+ = f$$

$$|f(x)| = |f|(x) = f(x)$$

$$\therefore \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2} = \frac{f(x) + f(x)}{2} = f(x) = f^+(x)$$

$$\text{If } f(x) < 0 \Rightarrow f^+(x) = 0, |f|(x) = -f(x)$$

$$\therefore \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2} = \frac{f(x) - f(x)}{2} = 0 = f^+(x)$$

$$\Rightarrow \frac{f + |f|}{2} = f^+ \quad (\text{proved})$$

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(39) Define Borel set.

Sol: A set 'E' is said to be a Borel set if it can be expressed as countable union or intersection of open or closed sets.

e.g. $\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ is Borel set.

(40) Define Sum Module 1 and Translation Module 1.

Sol: Sum Module 1:

Let $x, y \in A = [0, 1)$ and we define sum module 1 of x and y as

$$x \hat{+} y = \begin{cases} x+y & \text{if } x+y < 1 \\ x+y-1 & \text{if } x+y \geq 1 \end{cases}$$

Translation Module 1.

Let $E \subseteq A = [0, 1)$ and $y \in A$, we define a translation module 1 of "E" as

$$E \hat{+} y = \{x \hat{+} y, x \in E\}$$

Set of all Sum Module 1.